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Traders' Dilemmas and Development: A Variety of Solutions

RICHARD GRABOWSKI

Introduction

Economists tend to spend a great deal of time discussing how markets function, but precious little time on explaining how they are established. This gap in knowledge became apparent as many of the former socialist nations attempted to make the transition to a market economy. The same problem bedevils development economics. The efficient operation of an extensive national market system is considered essential for rapid economic development, but relatively little attention has been directed to trying to understand how such a system comes into being. The problem is analysed in this article in the form of the traders' dilemma.¹ Buyers and sellers meeting in the market are faced with the following problem. Should the buyer (seller) choose to behave in an honest, open and cooperative manner with the seller (buyer) or should s/he choose to behave uncooperatively and dishonestly? If either buyer or seller, or both, choose the latter, extensive trade between the two will not be likely to occur or, if it does, will be shortlived in nature. More importantly, one can show that under certain reasonable circumstances the uncooperative choice can be rational for both buyer and seller. Thus the question becomes: what sort of environment will allow a cooperative, honest solution to the traders' dilemma?

Section I of the article discusses the traders' dilemma in more detail within the context of some simple notions from game theory. Section II explores institutional aspects of the dilemma and Section III outlines a variety of different institutional solutions. Section IV uses the development experiences of Japan before World War II, South Korea and Southeast Asia as examples of the various institutional solutions to the traders' dilemma.

I

One can view the traders' dilemma as a form of the famous Prisoner's Dilemma. In this game there are two players (I, II) who each have a choice between two strategies (A, B). Strategy A represents cooperative behaviour while B represents uncooperative behaviour. The payoffs in this simple game are illustrated in Figure 1, with the first number representing the payoff to I and the second to II.

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	A	B
A	10, 10	0, 12
B	12, 0	2, 2

FIGURE 1. Outcomes of cooperative and uncooperative strategies in the Prisoner's Dilemma.

Presuming that the game is played only once and that the choices are made simultaneously, a seemingly odd result emerges. It is in each individual's self-interest to choose to be uncooperative, with the result being that both individuals choose B, which is an equilibrium solution to the game. As a result, the group ends up with a socially inferior solution where the rewards are (2, 2). Thus individuals pursuing their self-interest will achieve the worst possible situation for the group.

The above dilemma is descriptive of the situation faced by buyers and sellers seeking to trade. The seller could choose to be honest and provide a good quality product to the buyer at the agreed-upon terms. Alternatively, the seller could choose to dilute the quality of the product, take payment and not provide the product, or indulge in a whole host of other behaviours which represent uncooperative activities. In this same manner buyers can choose either to trade in good faith or choose to be uncooperative. Uncooperative behaviour could include taking the product without paying for it, for example. One might characterise this as a problem of trust between two partners in the process of trade.

The effect of such a dilemma is to raise the cost of exchange. This notion is captured in the literature on transaction costs.² Transaction costs include the costs of finding a trading partner, gathering information on potential trade partners and enforcing the terms of any agreement. Most of these costs are related to a lack of information and the fact that the gathering of such information is a costly process. However, the previous paragraph indicates that the lack of trust between potential trade partners would significantly increase the transaction costs of engaging in market exchange. The effect of such transaction costs is illustrated in Figure 2 which represents typical demand and supply curves for a market. With zero transaction costs the equilibrium would be at P_c and Q_c . Transaction costs drive a wedge between the price paid by buyers and the price received by sellers. In this figure the wedge is represented by distance AB. As can be seen, the larger the wedge (transaction costs) the less market exchange will occur. In the extreme case, no exchange will occur and the market will cease to exist.

Of course, the above analysis is much too simplistic and pessimistic in nature concerning the possibilities of successful trading. It presumes that the traders' dilemma game is played only once. If it is played an indefinite number of times and if the discount rate applied to the future is not too high, different results can emerge. Now a player who chooses to cheat (uncooperative choice) in a market exchange must weigh the short-run gain against a possibly large long-run loss. Specifically, the cheated player may retaliate by choosing to be uncooperative (cheat) in future plays of the game. In terms of Figure 1, Individual I must weigh

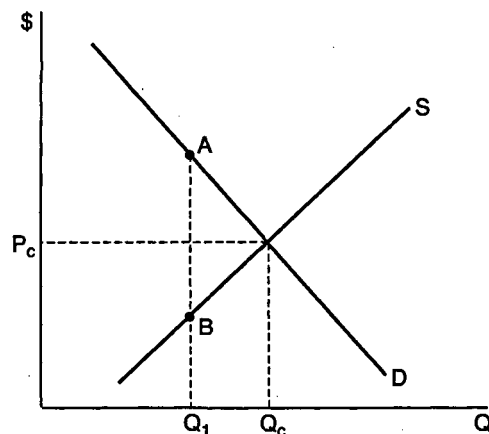


FIGURE 2. Effect of transaction costs on engagement in market exchange.

the short-run gain of cheating (return of 12 rather than 10) against future losses (return of 2 rather than 10) stemming from retaliation by his trading partner. Fear of such retaliation upon the part of both players results in cooperation becoming a possible equilibrium.

Although a step forward, the analysis requires that the same traders interact in the same market over a long period of time before the possibility of successful trade via markets becomes a possible equilibrium. However, as Kreps points out, this restrictive environment can be expanded somewhat by introducing the notion of reputation.³ If one assumes that somehow information concerning the past choices of all players is available to any particular player, then cooperative solutions will be an equilibrium in the game even when players do not interact continuously over a long period of time. In this situation dishonest behaviour (non-cooperation) by a player results in the player being designated as untrustworthy. Thus individuals must once again weigh the short-run gains of cheating against the long-run losses that come from developing a reputation for dishonesty. The losses could result either from retaliation by other players or the unwillingness of the latter to interact at all with those of untrustworthy reputations.

These ideas are similar to the notion of embeddedness which has been developed by economic sociologists such as Granovetter.⁴ He argues that trust between individuals can actually be produced. He stresses that this can happen through personal interaction among individuals in structures or networks of relationships which generate trust and discourage uncooperative behaviour. This occurs through the creation of personal reputations which act as mechanisms to produce trust. Economic activities such as exchange between traders are then embedded in these networks of social relationships. Platteau agrees with Granovetter that such social embeddedness can generate high levels of trust.⁵ However, such mechanisms will only work with groups characterised by dense networks of social relationships usually reinforced by ethnic, kin or regional ties. It is here that individuals will be able to access information on potential

exchange partners at relatively low cost. As the number of individuals and, therefore, the size of the group grows, the informational requirements are likely to become so extreme (costly) as to prevent embedded relationships from providing a solution to the traders' dilemma outlined previously. The implication is that reputational mechanisms working through embedded relationships are likely to work only within relatively small communities.

It follows from the above that, within small regions where social networks are dense, cooperative solutions to the traders' dilemma are likely to arise. However, between or among such regions the lack of such a social network is likely to imply that non-cooperative solutions dominate. *Within* these small regions market activity is likely to be embedded and quite intensive; *between* and *among* the regions markets are likely to be weak or nonexistent. Integration of regions via the market will not occur. These ideas fit nicely with Rothenberg's distinction between market-based exchange and a market system.⁶ Market-based exchange has existed throughout most of man's history and involves transactions between buyers and sellers in markets. However, there is no tendency for the prices of goods and services of similar quality in different regions to converge. Alternatively, market systems occur through the integration of regional markets with the consequence that prices for goods of similar quality converge. The creation of such a market system would seem to be crucial to the overall development process.

II

There are a number of solutions to the traders' dilemma, or market integration problem, outlined in the previous section. Some insight into these solutions is provided by the work of Robert Frank.⁷ He assumes that there is a large population of individuals and that everyone in the population is of two types, either a cooperator or non-cooperator. The cooperators refrain from cheating in dilemma situations, while the non-cooperators will cheat given the opportunity. Within this context Frank is interested in determining what will happen in a survival struggle between these two groups. It is assumed that the proportion of cooperators will increase as long as their rewards from playing the game exceed those of the non-cooperators and vice versa. The reward structure of the game will thus resemble that in Figure 1. Yet, if one assumes that there is no way to distinguish between cooperators and non-cooperators and that individuals are paired up randomly, then no matter what proportion of the population is made up of cooperators (as long as it is less than 100 per cent), non-cooperators will always reap higher rewards. In this situation, because of the relative success of the non-cooperators relative to cooperators, the latter will eventually disappear from the population. Equally, but alternatively, if one assumes that there is a mechanism for distinguishing cooperators from non-cooperators and that this information is made available to all players of the game, then the situation is altered dramatically. Cooperators will seek out cooperators and refuse to interact with non-cooperators. The latter will only be able to interact with fellow non-cooperators. Using payoffs again as illustrated in Figure 1, it follows that

cooperators would gain relative to non-cooperators, with the latter disappearing from the population.

Frank considers a third possibility. Suppose it is possible to distinguish between the two groups, but that this can only be accomplished by paying a price. In this situation, an equilibrium arises in which a proportion of the population remains cooperative while the rest are non-cooperative. This proportion depends on a number of factors, one of which is the cost of gaining information concerning the cooperators and non-cooperators. The lower this cost, *ceteris paribus*, the larger the proportion of the population made up of cooperators and vice versa. Applied to the problem under discussion here, this indicates that, for individuals in dense social networks connected via kin, ethnic or regional ties, the cost of information concerning potential partners in exchange is extremely low. Thus the traders' dilemma within this limited group is overcome as cooperators predominate in the population and market exchange thrives. Developing nations, especially the rural areas of such nations, are composed of large numbers of these groups.

Exchange between such islands of economic activity is, however, quite limited. This is because there is little information on how traders between two different groups will behave with respect to each other. The cost of providing such information is quite high. In the extreme case it would be impossible for members of one group to tell which members of another group would be cooperative and which non-cooperative. In this case non-cooperative choice will predominate in exchange relationships between groups and a market system will fail to develop. The great merit of Frank's analysis is that he pinpoints the crux of the problem preventing market integration. It is a lack of information about who is trustworthy and who is not which keeps market integration from occurring. The implication is that such integration will follow if the necessary information can be generated at a relatively low cost per transaction.

At this point in the article it is time to recognise that the traders' dilemma is really a game played among three players: buyers, sellers and intermediaries. The point is that buyers and sellers generally do not interact directly in the marketplace. Instead, the arranging of the exchange is usually left to intermediaries who in this article will now be identified as the traders. Now each trader faces a traders' dilemma game with each buyer and seller separately. This group's function is to gather information about the behaviour of potential buyers and sellers and use this information to arrange successful exchanges. In this sense one can view the production of information, concerning how cooperative or uncooperative particular buyers and sellers are, as the main business of traders.

One can formalise the above analysis a little by viewing each trader or merchant as offering prices to sellers and buyers with the return to the trader coming from the price differential. In an environment in which information on the past choices of buyers and sellers is not available, the trader is faced with a single-play traders' dilemma situation. From the transactions cost perspective, the costs of this type of exchange would probably be very high to the merchant, perhaps so high that he refuses to arrange transactions. However, there may be a technology available to provide low-cost information concerning the behaviour

of potential buyers and sellers. In small societies or groups this would involve a dense network of social relationships based on kinship and regional ties which bind people together. It is these social relationships which provide the information concerning the behaviour of potential buyers and sellers. Traders within a particular group can tap into this information system so as to be able to identify cooperators. In this environment, as Frank points out, cooperators will flourish and markets within the group will be organised by traders.

The method for generating information discussed above is labelled in this article as the traditional technology.⁸ Although it is highly effective in areas where social networks are strong and pervasive, it ceases to function between different regional or village groups. The social relationships linking individuals between groups are likely to be sparse. However, there is an alternative technology available for generating the information that traders need to operate effectively between groups. Clues to identifying this technology are provided in the work of Levy concerning the advantages which intermediaries (traders) have in arranging exchange between buyers and sellers.⁹ Each group characterised by dense networks of social relationships has a set of rules and regulations (written and unwritten) which govern trade within their particular group. These rules generally reflect the local culture. Before traders can arrange exchange between different groups these rules and regulations must be learned. Such learning involves large, indivisible set-up costs that, without trading intermediaries, would have to be borne by each buyer and seller (each from a different group). The advantage of a trader, then, is that these costs can be spread over a large number of potential buyers and sellers, thus significantly reducing information costs per transaction. This argument, of course, implies that the advantage of traders, relative to each individual buyer and seller arranging their own transactions, is found in increasing returns to scale in transactions.

There are, moreover, other sources of increasing returns. A second derives from the fact that, with respect to buyers, information about their needs for specific products will be required if sellers are to be able to meet that demand effectively. Buyers seeking to purchase a particular product require information about the attributes of the goods produced by particular suppliers. The gathering of this information would involve significant expense upon the part of individual buyers and sellers, but an intermediary can spread these costs over a large number of buyers and sellers, again significantly reducing costs per transaction.

The third and most important source of increasing returns relates to the gathering of information concerning the reputation of particular buyers and sellers. The traditional mechanism for generating such information involves using kinship, regional or cultural ties and is limited to small groups. However, traders behaving as entrepreneurs could seek to establish links between buyers and sellers of different groups, disregarding kinship, regional or cultural ties. This would involve gathering information on the behaviour of buyers and sellers concerning the identity of the cooperators and non-cooperators. In other words, traders would have to gather information on the reputations of buyers and sellers in different groups. This would involve creating an extended network between groups and the initial costs of doing so are likely to be great. There is, however, a complication that may occur.¹⁰ Individuals who engage in transactions outside

the group may be subject to retaliation from other members of the group. Thus the cost to buyers and sellers of engaging in such across-group exchange via a trader will be increased. To make such trade across groups attractive to buyers and sellers, traders will have to offer sellers a higher price and buyers a lower price than they could receive within their respective groups. This is an additional barrier to such exchange in addition to the high costs of gathering information discussed above.

One may view the information costs as start-up or overhead costs. Once the information on reputation is generated and is in the hands of the trader, it can be used over and over again in the selection of proper buyers (honest cooperators) for proper sellers (honest cooperators) for exchange involving a wide variety of goods. In other words, the intermediary who bears the cost of collecting such information would be able to spread this fixed information cost over a large number of transactions, thus dramatically reducing the cost per transaction. Thus intermediaries would be able to reap increasing returns to the gathering of information. As to the retaliation costs that buyers and sellers may have to bear as a result of engaging in exchanges across groups, these too are likely to decline as the number of such transactions increase. As a larger proportion of each group becomes involved in such trade, retaliation is less likely to occur. Thus economies of scale with respect to these costs also occur.

A fourth and final source of increasing returns is related to the financing, warehousing and transportation costs involved in carrying out exchange between buyers and sellers. The various facilities used to carry out exchange for one type of good can be used for other types of goods as well. In other words, traders can also reap economies of scope when dealing with a wide variety of different products. While these are economies of scope when one thinks of the output in goods and services, they are economies of scale when the output being considered is the number of transactions. Being able to use financial and transportation facilities for a wide variety of goods and services dramatically reduces the cost per transaction.

The increasing returns to scale in terms of volume of transactions are crucial to making a distinction between the traditional mechanism or technology and what I have called the modern technology for generating information allowing the identification of buyers and sellers who are cooperators. The traditional technology requires or relies upon a dense social network among kin or utilises regional or cultural links to generate the information necessary to identify cooperators. Reputations generated this way allow for intensive market exchange among members of the group. However, beyond the boundaries of the group this mechanism fails to function and thus market exchange rarely evolves into a national market system. The modern technology for generating information requires traders (intermediaries) to establish links between buyers and sellers in different groups and is the subject of significant economies of scale, thereby increasing returns. In sum, one may visualise the relationships in the following way. For small levels of transactions the costs per unit from utilising the traditional means for generating information are much lower than for the modern technology. However, as the volume of transactions increases the modern technology for gathering information on cooperative behaviour (reputations) of

buyers and sellers generates significant increasing returns. At a large enough scale the costs per unit of utilising this technology will drop below that of the traditional technology.

This analysis can now be related to understanding how market systems can emerge out of market exchange. It is in the rural areas that markets are most fragmented and in which islands of intense economic activity are only loosely linked, if at all, to each other. Further, for most developing nations it is agricultural production which involves the bulk of the population. In particular, the production of food grains for subsistence purposes generally dominates the activities of farm households. Thus it is in the agricultural sector, involving farm families producing for subsistence, in which the potential is greatest in terms of numbers of sellers that could be involved in market exchange. In terms of buyers, again it is food grain purchases which take up the largest proportion of the budgets of the poor. Their purchases are of course significantly restrained by their limited budget. However, the potential in terms of numbers of buyers and transactions is probably greatest for purchases of food. The point here is that the greatest potential market (numbers of buyers, sellers and transactions) in many less developed countries is the market for basic food grains. I say potential market because in most instances limitations on both the buyer and seller side limit the actual size of the market (number of transactions).

The activation of this large potential market depends on the willingness of traders to move beyond the traditional means of collecting information on reputations through kinship ties, extended family, cultural links or regional ties. It requires these intermediaries to reach beyond their own group to establish links with other groups. This is the modern technology for gathering the information. However, here is the rub. For small levels of transactions the costs of the modern approach are much higher than using the traditional technology. In other words, enough buyers and sellers must commit themselves to market purchases and sales if intermediaries are to switch successfully to modern information gathering. This is the classic case of a coordination problem.

Coordination problems played an important role in early theorising on the problems of economic development. Rosenstein-Rodan argued that the expansion of a single manufacturing industry by itself is likely to fail in a less developed country.¹¹ For example, a shoe factory established in isolation will certainly generate enough income to buy all of the shoes produced. However, this income is not likely to be spent entirely on shoes. Instead, much of the demand would spill over into other commodities and thus the shoe factory would probably fail. However, if a number of investments in different manufacturing fields are carried out simultaneously, then each will provide markets for the production of other sectors. More to the point, investors in potential shoe production hesitate to invest for fear of a lack of market, investors in radio production hesitate for fear of a lack of market and so on. If the investment plans of all these different industries could be coordinated in the form of a big push, they will all be more likely to succeed. Murphy, Schliefer & Vishny have formalised this Big Push analysis within the context of imperfect competition with demand spillovers (pecuniary externalities).¹² Within this structure there are two possible equilibria, one with no industrialisation (cottage manufacturing)

and one with industrialisation (utilising increasing returns to scale technology). The ability to move from the cottage manufacturing equilibrium to the industrialisation equilibrium depends again on the coordination of investment plans.

The same sort of problem arises in the context of the argument of this article. Agricultural producers (suppliers) interact with intermediaries (traders). These producers are willing to undertake significant investment in the resources and techniques necessary to expand production for sale rapidly if they can be assured that there will be a market for such output that allows them to profit from the investment. In turn, the traders are willing to invest in the modern technology for the gathering of information concerning the reputation of sellers, but only if the number of transactions undertaken by sellers is sufficiently large to allow the trader to take advantage of the economies of scale connected with the modern technology. On the other side, buyers will also have to restructure their behaviour significantly to participate extensively in market purchases. They will have to switch from a subsistence mode to specialisation in production and the reallocation of resources that comes with this. They would then purchase the bulk of their food in the market. However, their ability to benefit from this restructuring is dependent upon the extent to which a reliable market for food and other agricultural goods exists. This in turn depends upon the ability of traders to assure the buyers an adequate supply of goods and this in return requires the traders to utilise the modern technology for gathering information concerning buyers' reputations. Again, this is only possible if enough buyers participate to allow traders to take advantage of the economies of scale from the modern technology for generating knowledge.

III

Solutions to this problem require coordination of the activities of buyers, traders and sellers. Obviously there is a potential role for the state to play here. It could seek to play an active role in this coordination process. However, one must be careful not to see the solution in terms of extremes (either the government solves the problem or it does not get solved). The existence of the coordination problem indicates that there are significant profitable opportunities available to any group that is capable of resolving this problem even partially. Private entrepreneurs may seek to capture these profits by extending social networks as intermediate traders and undertaking the coordination of investment. The governmental solution is the other extreme possibility which has the government creating networks linking groups and coordinating investment by buyers and sellers. However, between the two there sit various mixtures of private and public entrepreneurial coordinating activity. The actual mixtures utilised in various countries will differ dramatically, with this difference linked, to some extent, to initial conditions.

Three possible solutions to the coordination problem can be outlined. The first represents a broad mixture of public and private coordination. In countries which are importing food grains, this importation serves as a barrier to prevent the creation of an integrated national market. Because a large part of the food grain needs of buyers is being met via imports, producers will find opportunities for

marketing increased output dramatically curtailed. They are thus likely to invest less in production enhancing activities beyond that necessary for the provision of subsistence levels. Buyers are meeting much of their needs via imports and are unlikely to interact with domestic traders. The latter, finding the actual market size small (limited transactions), conclude that it is unprofitable to shift to the modern technology for information gathering (subject to increasing returns to scale).

A mixture of private and public coordination would then involve the following. First, by the state protecting the agricultural sector, domestic buyers are given incentives to expand their purchases of domestically produced food grains. They are in a sense being pushed into buying from domestic producers. Second, investment by the state into productivity enhancing techniques for rural producers will assure the ability of domestic producers to respond to the increased demand. Private entrepreneurs seeing the signals emanating from government policy are thus more likely to see and seek profitable opportunities resulting from investing in the expansion of the social network which serves to link groups into market systems. They are accordingly likely to invest in the creation of modern technologies for information gathering which are subject to increasing returns to scale.

Note that the above solution to the traders' dilemma is dependent upon the expansion of the productivity of the bulk of the producers in the rural sector, generally food grain producers. It is this expansion of production combined with protection which involves increasing numbers of buyers, sellers and traders in rapidly growing numbers of transactions which in turn trigger the increasing returns attached to modern mechanisms for gathering information on the cooperativeness of potential buyers and sellers. Thus the market or, I should say, the market system, is the result of growing interaction by the various players of the game, spurred on by the coordinating activities of private entrepreneurs and the government. More plainly, agriculture serves as the birthplace of the market system. The national market resulting from this interactive process and its rapid growth would provide the basis for successful industrialisation, as domestic manufacturing firms would find profitable opportunities for trying to meet the increasingly diversified demand from the agricultural sector for manufactured goods. This would be an industrialisation process driven by domestic demand which in turn was driven by agricultural growth.

An alternative solution to the traders' dilemma is suggested by the mirror image of the above process. Specifically, how might the traders' dilemma be solved in an environment characterised by the slow growth, or perhaps the stagnation, of productivity among the bulk of agricultural households? In such a situation the bulk of the food needs will be met through imports. Networks connecting self-sufficient groups within the rural sector are unlikely to develop, further retarding specialisation and productivity growth. A clue as to how the traders' dilemma might be resolved in this situation is provided by the recent work of Langlois & Robertson,¹³ as well as that of Granovetter¹⁴ and Feld.¹⁵ Langlois & Robertson view firms and other types of organisations as being made up of two components: an intrinsic core and ancillary capabilities. The intrinsic core 'comprises elements that are idiosyncratically synergistic, inimitable, and

noncontestable'.¹⁶ The ancillary capabilities are contestable capabilities which could be purchased from other sources. What is intriguing about their analysis of firms and organisations is their description of intrinsic core activities. They describe these as being part of the organisational culture of the firm. This culture is made up of the habits and routines that individuals acquire through practice. They are unique to the firm and 'constitute the essence of the firm in the short and medium terms'.¹⁷

Compare this to Williamson's discussion of the boundaries of the firm.¹⁸ The boundary of the firm concerns those transactions that take place via the market and thus occur between firms, as opposed to those transactions which are conducted within the hierarchical firm. Of course, the side on which particular transactions fall—within the firm or in the market—depends on relative transaction costs. What is important here is the advantage Williamson attributes to organising transactions within the hierarchical firm. The hierarchical organisation of the firm can be used to tame opportunism upon the part of individuals. This opportunism involves pursuit of self-interest even when it results in an inferior outcome for the group (firm). This is in fact the traders' dilemma discussed previously. In those situations in which individuals tend to cheat by choosing to be uncooperative, the resulting equilibrium can be inferior for the group as illustrated in Figure 1. In this situation these sorts of transactions would, according to Williamson, be better handled within the confines of the firm where the authority of hierarchical relations can be used to rein in the pursuit of self-interest. As can be seen, there is a subtle difference between the analysis of Langlois & Robertson and that of Williamson. The former refer to the core of the firm as a set of habits and expected behaviours involving reciprocal relationships that have come to be viewed as accepted behaviour and are, in the short and long run, unique to that particular firm. Williamson appeals to authority stemming from hierarchical relations of power. The problem with this is that it is itself subject to the corrosive effects of opportunistic behaviour. By comparison, the reciprocal patterns of behaviour outlined by Langlois & Robertson rely less on authority and more on interpersonal, reputational mechanisms.

This is related directly to the work of Feld, who attempts to explain the patterns found in social networks. In particular, he is interested in explaining clusters of social networks and the connections between such clusters. He argues that such clusters tend to evolve around foci. 'A focus is defined as a social, psychological, legal, or physical entity around which joint activities are organized (e.g., workplaces, volunteer organizations, hangouts, families, etc.)'.¹⁹ It follows, then, that as a 'consequence of interaction associated with their joint activities, individuals whose activities are organized around the same focus will tend to be interpersonally tied and form a cluster'.²⁰ The implication is that firms can serve as the focus around which dense social networks will arise through continuous interaction of the individuals involved. Thus the core of the firm becomes this dense social network tying a group of individuals together. Within this core cooperative solutions to the traders' dilemma are created via continuous social interaction.

One can briefly summarise the analysis. In those environments in which social

networks are not well developed, they can be created within the context of the firm. Thus the lack of markets caused by the widespread existence of the traders' dilemma is overcome by substituting the firm for the market. The firm serves as a focus around or within which a social network can be created. The government's role here would be to promote economic development not via the creation of a national market system, but by circumventing the lack of such a system through the establishment of large business conglomerates within which a core set of social networks would evolve within the firm. The government's job of coordinating overall expansion is thereby simplified, since only the activities of a small number of large firms need to be coordinated. This solution to the traders' dilemma in effect forces repeated plays of the dilemma within the context of the firm and results in the reputational mechanism serving to assure cooperation within the firm. The firm and its core of developing social networks substitutes for the market.²¹

There is still a third path by which the traders' dilemma can be solved. Remember that the dilemma concerns the inability of traditional systems of information-gathering to integrate groups into a market system. The first solution involved a coordinated expansion in agricultural production (of a widely produced agricultural good) combined with protection of domestic agricultural production. These signals, triggered by government policy, would indicate to domestic traders that significant increases in transactions by sellers and buyers of these agricultural goods will occur. This would allow domestic traders to invest in modern techniques of information gathering which are subject to significant economies of scale. The key to the success of the above process revolves around productivity growth in the agricultural sector and coordination via protection of this sector by the state. However, those nations having a comparative advantage in agricultural goods produced by the bulk of the population will likely find the expansion of trade in these agricultural items can serve to activate the development of an integrated market system. The export expansion would greatly increase the number of potential transactors, thus indicating to traders the profitability of utilising expanded networks of relationships to identify producers that are cooperators as well as buyers that are cooperators.

In this scenario, though, one point must be kept in mind. If the export crop is grown only by a small proportion of the rural population, the number of transactions involved is likely to be small. This would pose an obstacle to the use of modern information-gathering mechanisms by traders. As a result, the development of an extensive system of markets is likely to be limited, or fragmented and undeveloped. In short, if export growth is to serve as an activator of market integration it must involve a commodity produced by the bulk of the population.

Historically, the network of traders discussed above is often ethnically based. That is, most of the trading operations are handled by an ethnic group, usually a minority, within the nation. Given restrictions which limit such groups' access to land in the agricultural sector, their activities are often concentrated in commercial sectors. Thus a pre-existing dense network of social ties allows the reputational mechanism to be effective in conveying information concerning

whether buyers and/or sellers are cooperators or non-cooperators. This dramatically reduces the cost of conveying the information necessary for the integration of markets. Of course, one must point out that the existence of such ethnic groups often creates tensions within the society between buyers and sellers from one ethnic group and traders from another. Ethnic conflict within such a situation would fragment the market and delay the overall development process. However, if such tensions can be contained, ethnic ties become a useful mechanism for drastically reducing the cost of information flows.

Moreover, there is another advantage stemming from ethnic trading groups. The coordination role of the state may be partially subsumed within the activities of such ethnic groups. That is, ethnic ties can serve as a conduit through which credit, technical advice and other critical elements in the development process can be allocated. Often in developing countries the efficient allocation of such resources is limited because these types of inputs generate significant positive externalities. Individual investors who ignore such externalities are likely to underinvest in such crucial inputs. Ethnic ties serve to somewhat modify this situation by at least partially internalising the externalities. Thus extended ethnic ties take on a coordination function which, in many other countries, the state must undertake.

The three solutions to the traders' dilemma discussed in this section are, of course, not the only possibilities. However, they represent major types which I believe characterise the process of development in much of East and Southeast Asia. In the next section examples will be drawn from these areas which illustrate the three solutions.

IV

One of the best examples of the first type of solution is provided by the early development experience of Japan, by which is meant the late Tokugawa (late 18th and first half of the 19th century) as well as the Meiji period (latter half of the 19th and early twentieth century). It was during this time that the market system became established, Japan rapidly industrialised and the foundations for rapid postwar development were laid. Remember that the first solution to the traders' dilemma involved protection of the agricultural commodity produced by the bulk of the rural households and an allocation of resources to farms so as to promote increased productivity. During the latter part of the Tokugawa period both elements of the first solution occurred, though not as the result of purposeful government policy. The point is that, throughout the Tokugawa period, Japan followed an isolationist policy with respect to the rest of the world. Foreign-produced agricultural goods, in particular rice, could not be imported and thus the agricultural sector was protected from competition from outside. One might think that such isolation would have harmed Japanese agriculture, since it did not allow rice to be exported. However, the Meiji experience shows that once Japanese markets were opened to trade, the initial comparative advantage in rice quickly disappeared. In effect, this isolationist policy protected Japanese rice farmers and forced domestic consumers to purchase only domestically produced rice.

In addition to the above, villages were increasingly able to resist tax increases imposed upon them by feudal lords and the Tokugawa Shogunate. This was related to the fact that the warrior group had been previously removed from the countryside. The villages not only took on responsibility for collecting taxes but they also became mechanisms for resisting tax increases. As a result, during the latter half of the Tokugawa period land tax rates and assessed land values did not rise.²² The end result was that any increases in output and income remained in the rural sector. This provided not only the incentive to innovate, but also, to some extent, the means, since the increased returns could be reinvested to raise agricultural productivity.

According to the analysis offered in this article one would expect traders to find it profitable to seek to integrate isolated market activities into market systems. As the number of buyers and sellers increases dramatically, it would become profitable for traders to invest in creating the linkages necessary to provide information concerning buyers and sellers who were cooperators. In this manner, cooperators would come to dominate and a national market be created. Indeed, in Japan during this time merchant houses engaged in trade arose first in rice and other basic foodstuffs and national markets developed for major crops such as rice and silk. However, since the facilities necessary for carrying out trade in basic commodities can, once established, be used to organise markets in other commodities, market systems also began to develop in cotton, sugar, indigo, speciality fruits and vegetables.²³

The state played a more purposive role in this process in the emerging Meiji period. During the Russo-Japanese War (1904–5) a tariff was levied on imported rice and was made permanent in 1906. Political conflict over the tariff eventually resulted in rice from Formosa and Korea (colonies of Japan) being made duty free. The net effect was to raise rice prices in Japan, Korea, and Formosa above world levels.²⁴ In addition, the new government reformed the land tax, reducing it and converting it from an in-kind to a cash basis. This acted as a further spur to suppliers to produce for the market. Also, the government devoted additional resources towards agricultural research and extension and land infrastructure and output growth rose to 2.5 per cent during the period 1904 to 1908.

The solution to the traders' dilemma in the Japanese context resulted in the creation of national markets for agricultural goods, increased agricultural productivity and a growing demand for relatively simple manufactured goods. True, large conglomerates existed, but they generally produced raw materials or intermediate products which were then processed by the small firm sector into final products. The new networks created via trading groups within agriculture served as the mechanism for establishing national markets in these types of simple manufactured goods. However, the key to long-run successful development still depended upon the ability to penetrate foreign markets via exports.²⁵

Once again, Japan was faced with a traders' dilemma. Outward-orientated growth required that individual firms in most new industries be willing to make significant investments in new technology. Countries with relatively abundant labour such as Japan seemed to have a natural comparative advantage in labour-intensive goods; nevertheless, the commodities produced still needed to represent the styles desired by foreigners, be produced at an acceptable level of

quality and be available in reliable, adequate quantities. The technology necessary to accomplish such goals was always likely to involve significant fixed costs. Traders seeking to play the intermediary role in marketing the product to foreign countries had to contemplate making significant investments in an information-gathering network that will seek to determine who among the domestic sellers and foreign buyers were cooperators. The establishment of such an international network was, like the domestic network, subject to significant economies of scale. The upshot was another coordination problem. Domestic producers were reluctant to invest in new technology necessary for exports as long as they were uncertain whether an international network for penetrating foreign markets was available. Alternatively, domestic trading groups were reluctant to create an international marketing network as long as they were uncertain whether domestic producers would make the required investments in new technology.

The solution to this problem involved a subtle bit of government coordination. However, its efforts were, once again, not directly aimed at coordinating the development of an international marketing network, but were instead a side effect. They involved subsidising the Japanese shipping industry and providing ready access to cheap credit for Japanese trading companies involved in overseas trade. The companies used their access to cheap credit and transportation to keep the cost of imported machinery and technology low—and then in turn used this as a mechanism by which to make modern technologies available to large numbers of very small domestic producers at low cost. Perhaps the best example of this process is provided by the growth and development of the cotton textile industry which, by means of the activities of trading companies, was able to penetrate foreign markets successfully.

An example of the second type of solution to the traders' dilemma is provided by the experience of South Korea. The previous discussion of the second solution was in the context of an agricultural sector where productivity growth was slow and there was, as a result, little development of a social network within the rural area. However, the firm serves as a mechanism within which such networks can be created. Individuals are gathered within the borders of the firm and are involved in repeated replays of the traders' dilemma game. Over time the reputational mechanism implicit in such activities will generate a set of social relationships within which the traders' dilemma is resolved. The state's role in this process is to set the overall economic goals (growth, innovation, foreign market penetration) and coordinate the activities of a small number of large firms aimed at achieving these goals. It must also possess the ability to discipline those firms where performance is poor while rewarding those whose performance is good (relative to the goal).²⁶

This broadly represents the process of development as it unfolded in South Korea. Initially, the Japanese wanted to use Korea and Formosa as sources of rice to supplement its domestic production. However, at colonisation in 1910 Korea was much less developed than Formosa. As a result, efforts to raise agricultural productivity were much less successful. Even more importantly, much if not all of the increased production was extracted for export directly to Japan. All the transactions in this trade were basically in Japanese hands. This implies that at independence (after World War II) extensive networks of social

relationships had yet to be developed and thus, for the most part, the Korean market was fragmented.²⁷ Market exchange occurred, but an integrated market did not exist.

However, according to Biggs & Levy, the South Korean state has been pursuing a government-directed learning strategy of economic development since the end of World War II.²⁸ Under this approach the government selected individual firms where the potential magnitude of dynamic internal economies was substantial. In addition, the state used a series of instruments to oblige them to expand and secure improvements in productivity. The learning refers to mastering technologies which were often imported from other countries. Indeed, technological learning was crucial for the process of development. However, the emphasis in this argument is on a different type of learning. Production within South Korea has been dominated by the establishment and evolution of a number of extremely large firms. As a result, many of the transactions which in other economies have occurred within the context of markets, occurred within the firm. The learning that was crucial here involved the establishment of social networks at the core of these firms, and the solving of the traders' dilemma within the context of the large firms. As a result, trading companies, subcontracting and other sorts of activities which occurred in markets embedded in social relationships in Japan came to be embedded within the firm in South Korea. The state's main function was to guide and coordinate the activities of these firms. This of course highlights a major difference between this and the first solution to the traders' dilemma. The former places a significant direct burden on the state to create the structure of large firms and coordinate their successful development. If this is to occur, it requires the state to be strong enough to resist efforts by these large firms to engage in rent-seeking rather than productivity-enhancing activities. In other words, this solution seems to be much more demanding on the capabilities of the state than the first.

The final solution is best exemplified by the economic development which is taking place in Southeast Asia. Between 1965 and 1980 the gross domestic product of Thailand, Malaysia and Indonesia grew at average annual rates of 7.39 per cent, 7.37 per cent and 7 per cent respectively; between 1980 and 1990 these growth rates were 7.6 per cent, 5.3 per cent and 5.4 per cent respectively.²⁹ Unlike South Korea, the state in all three countries seems to have been relatively weak, which suggests that the state-directed learning process does not provide a very good explanation of the success that has been achieved. On the other hand, part of the explanation is certainly to be found in the rapid growth and development which occurred in the agricultural sectors of these nations. During the 1960s product per worker in agriculture in Malaysia, Thailand and Indonesia grew at rates of 6.19 per cent, 4.39 per cent and 1.99 per cent respectively, whilst in the 1970s these growth rates were 4.1 per cent, 5.7 per cent and 2.3 per cent. Much of this agricultural growth was export-driven and brought about the development of a network of social relations supporting the development of market systems in all three nations.

However, the distinctive characteristic of Southeast Asian growth is its ethnic nature. The success of traders and entrepreneurial groups in solving traders' dilemmas can be linked to what Bates has called the soft factors of community,

symbolism and trust.³⁰ Rapid agricultural growth stimulated by export expansion would certainly provide opportunities for traders and entrepreneurs to invest profitably in expanding social networks, which would undergird market integration. However, symbolic and communal links that connect individuals in ethnic groups dramatically reduce the cost of establishing this link (for generating information on who the cooperators and non-cooperators are). In Southeast Asia the ethnic and communal ties among the Chinese have certainly made solving traders' dilemmas easier. In addition, ethnic ties have served as a mechanism for dramatically reducing the costs of establishing networks among firms within the manufacturing sector. These networks have evolved into what are commonly called business groups. Such groups are multicompany firms which transact in different markets, but do so under common entrepreneurial and/or financial control. Thus the investment strategy of these groups involves portfolio balance through significant diversification. These groups have predominantly been staffed at the higher levels by individuals linked through ethnic ties, i.e. Chinese.³¹ There are also formal business associations and informal social networks, dominated by ethnic Chinese, which serve as further mechanisms for overcoming traders' dilemmas.

This last solution to the traders' dilemma is probably the least demanding in terms of state direction and coordination. However, the difficulty in terms of drawing lessons for overall development policy for other nations concerns the role of such ethnic groups. First, a development process in which ethnic groups are an essential ingredient in the solution of the traders' dilemma is a fragile one. The tensions between ethnic Chinese and indigenous ethnic groups threatens to derail the process of overall development in parts of Southeast Asia. Second, ethnic groups capable of solving traders' dilemmas are not readily available for most nations. Equally, with the second solution to the traders' dilemma—government-directed learning within the context of large firms—another significant problem presents itself. Most developing nations lack a state that is strong enough to carry out such a programme. The lack of strength is of two types: first, the ability to use rewards and punishments to direct firms towards productive learning rather than rent-seeking; and, second, the ability to resist attempts by such large firms to capture the policy-making apparatus. Thus this solution is probably beyond the capabilities of most states. This means that the first solution is a more likely possibility for many of today's developing nations. Protection of agricultural production involving the bulk of the rural population and investment in the agricultural sector is probably less demanding, in terms of the state's ability, than the state-directed learning solution. However, this is not meant to imply that this path would be easy. It is merely that, given the circumstances facing most less developed nations, it is the one with the highest probability of success.

Notes

1. The traders' dilemma referred to here is substantially different from that referred to in Hans-Dieter Evans & Heike Schrader (Eds), *The Moral Economy of Trade: Ethnicity and Developing Markets* (Routledge, 1994).

2. For a discussion of transaction costs, see Douglass North, *Institutions, Institutional Change and Economic Performance* (Cambridge University Press, 1990).
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7. Robert H. Frank, *Passions Within Reason: The Strategic Role of Emotions* (W. W. Norton, 1988).
8. This concept is taken from Richard Grabowski, 'Agriculture, markets, and economic development', unpublished manuscript, undated.
9. Brian Levy, *Export Traders, Market Development, and Industrial Expansion*, Research Memorandum Series, Center for Development Economics, Williams College, Williamstown, MA, March 1989.
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13. Richard Langlois & Paul Robertson, *Firms, Markets, and Economic Change: A Dynamic Theory of Business Institutions* (Routledge, 1995).
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15. Scott Feld, 'The Focused Organization of Social Ties', *American Journal of Sociology*, Vol. 86 (1981), pp. 1015–35.
16. Langlois & Robertson, *Firms, Markets, and Economic Change*, p. 7.
17. *Ibid.*, p. 16.
18. Oliver Williamson, *Markets and Hierarchies* (Free Press, 1975).
19. Feld, 'The Focused Organization of Social Ties', p. 1016.
20. *Ibid.*
21. A similar idea is proposed in Brian Levy, 'Transaction Costs, the Size of Firms, and Industrial Policy: Lessons from a Comparative Case Study of the Footwear Industry in Korea and Taiwan', *Journal of Development Economics*, Vol. 34 (1991), pp. 151–78.
22. T. C. Smith, *Native Sources of Japanese Industrialization, 1750–1920* (University of California Press, 1988), pp. 15–49.
23. Penelope Francks, *Japanese Economic Development: Theory and Practice* (Routledge, 1992), p. 135.
24. Kym Anderson, 'Growth of Agricultural Protection in East Asia', *Food Policy*, Vol. 8 (1983), pp. 327–36.
25. See Richard Grabowski, 'Prewar industrialization in Japan', unpublished manuscript, undated.
26. Alice Amsden, 'A theory of government intervention in late industrialization', in: Louis Putterman & Dietrich Rueschemeyer (Eds), *State and Market Development: Synergy or Rivalry* (Lynne Rienner, 1992), pp. 53–84.
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28. Tyler Biggs & Brian Levy, 'Strategy interventions and the political economy of industrial policy in developing countries', in: Dwight Perkins & Michael Roemer (Eds), *Reforming Economic Systems in Developing Countries* (Harvard Institute for International Development, 1991), pp. 366–401.
29. Andrew MacIntyre, 'Business, government and development: Northeast and Southeast Asian comparisons', in: MacIntyre (Ed.), *Business and Government in Industrializing Asia* (Cornell University Press, 1994).
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